

# Battery Interfaces: Fundamentals, Sample Preparation and Analysis

**Lénaïc Madec**

Chargé de recherche CNRS, Section 15 - CNRS-IMN

[lenaic.madec@cnrs.fr](mailto:lenaic.madec@cnrs.fr)

# Outline

1° Context



2° The SEI formation and its properties



3° Sample preparation and analysis

# The use of batteries

=> They are very versatile so that they are used everywhere

## ELECTRONICS

Powering a wide range of everyday devices



Typical capacity:  
1 – 50 Wh

## PORTABLE COMPUTERS

Longer autonomy for work and study



Typical capacity:  
30 – 100 Wh

## TRANSPORTATION

Electrifying mobility for a sustainable future



| E-bikes / Scooters | Cars         | Buses / Trucks |
|--------------------|--------------|----------------|
| 300 – 700 Wh       | 40 – 100 kWh | 100 – 600 kWh  |

## PHONES

Always connected, always powered



Typical capacity:  
10 – 25 Wh

## BATTERIES

Powering our everyday life



## STATIONARY STORAGE

Storing energy for a reliable and resilient grid



Typical capacity:  
5 kWh – 100+ MWh

**mWh** milliwatt-hour  
(10<sup>-3</sup> Wh)

**Wh** watt-hour  
(10<sup>0</sup> Wh)

**kWh** kilowatt-hour  
(10<sup>3</sup> Wh)

**MWh** megawatt-hour  
(10<sup>6</sup> Wh)

From small (mWh) to large-scale (MWh):  
batteries power every scale of our modern life.



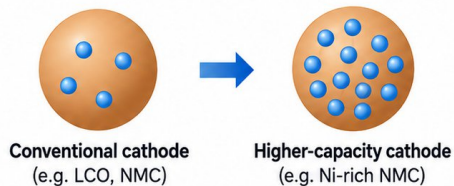
## Thanks to :

- High energy density
- Low self-discharge
- Long lifetime
- Wide temperature range
- Highly reliable
- Low maintenance

# Strategies to improve the energy

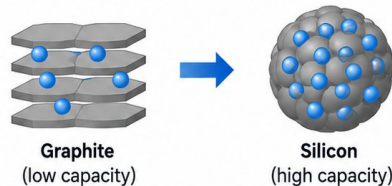
$$\Rightarrow \text{Energy} = \text{Capacity} \times \text{Voltage}_{\text{average}}$$

At the **positive** electrode (cathode)

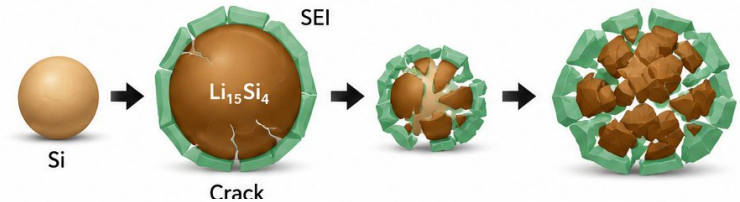


More Li stored

At the **negative** electrode (anode)

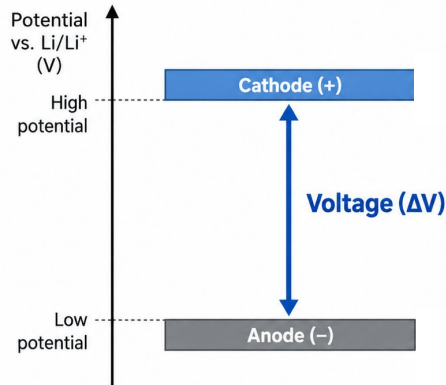


More Li stored

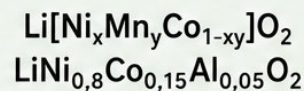


**! Huge volume expansion (> 200%)**

- $\Leftrightarrow$  material pulverization
- $\Leftrightarrow$  loss of electrical contacts
- $\Leftrightarrow$  huge electrolyte degradation (SEI)



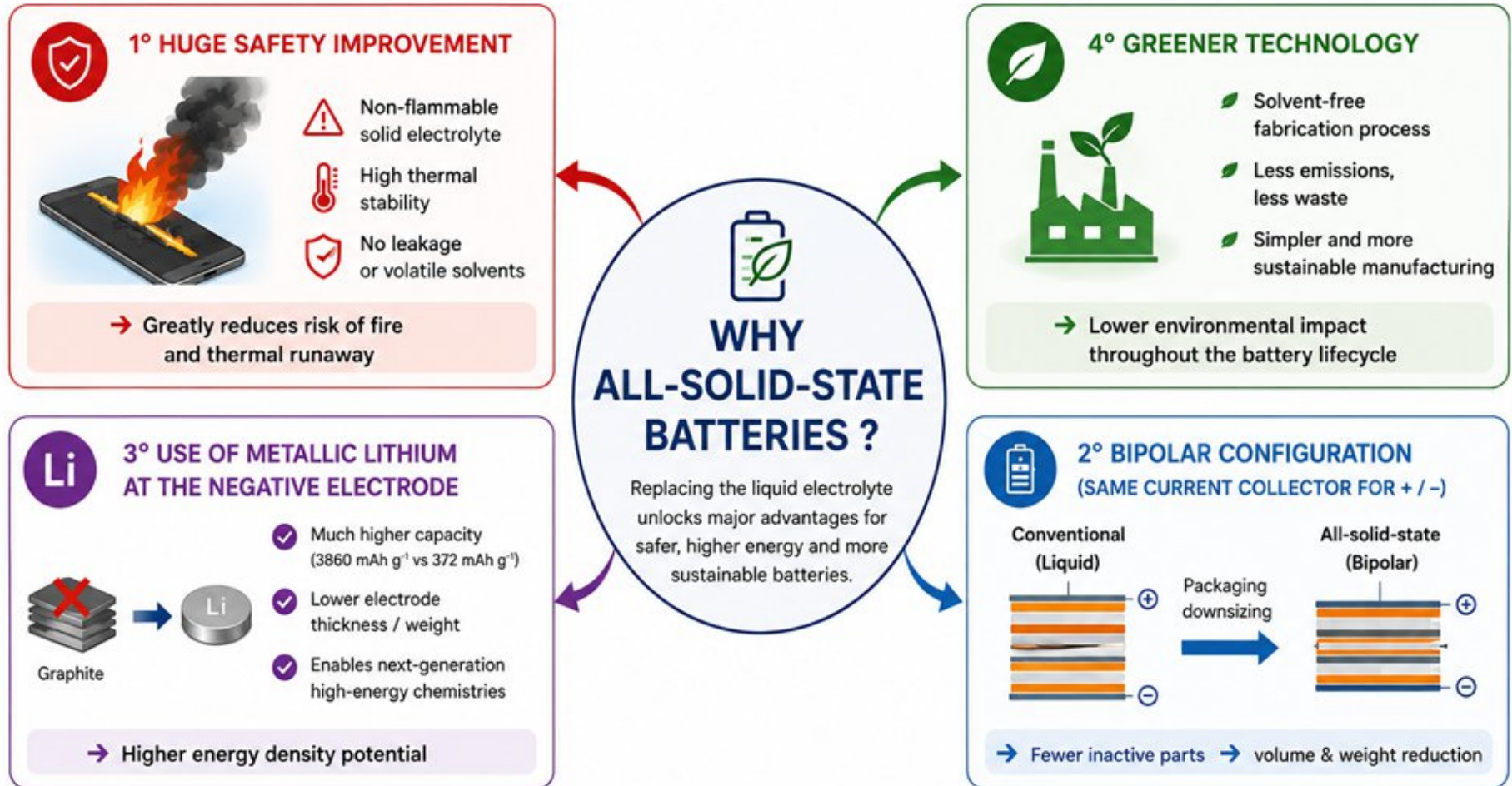
**Materials already available (up to 4.7 V)**



**HUGE electrolyte degradation (SEI)**

- $\Leftrightarrow$  impedance increase
- $\Leftrightarrow$  cell capacity decrease
- $\Leftrightarrow$  security issue (gas)

# Solid-state batteries

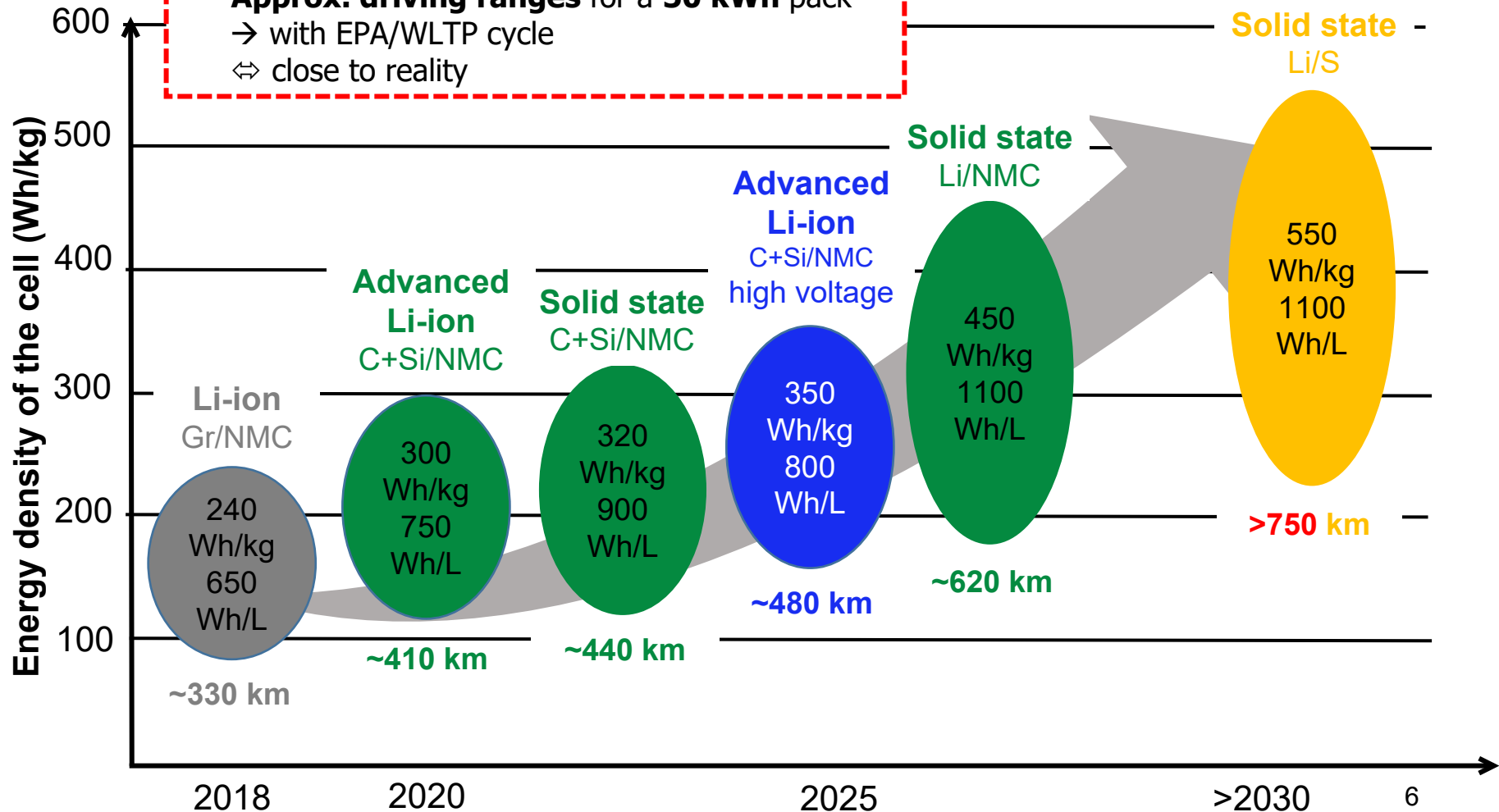


# Strategies to improve the energy

$$\Rightarrow \text{Energy} = \text{Capacity} \times \text{Voltage}_{\text{average}}$$

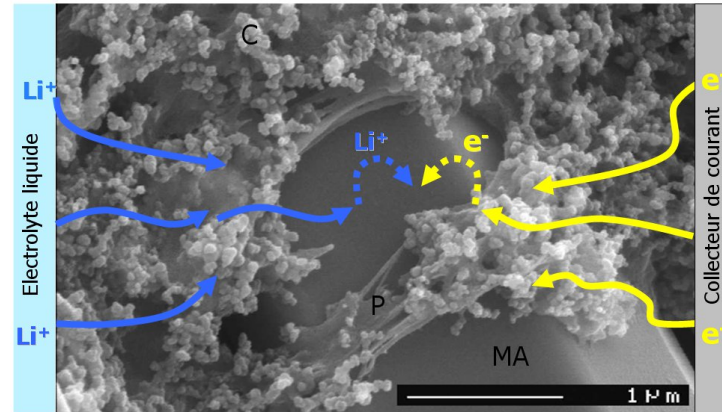
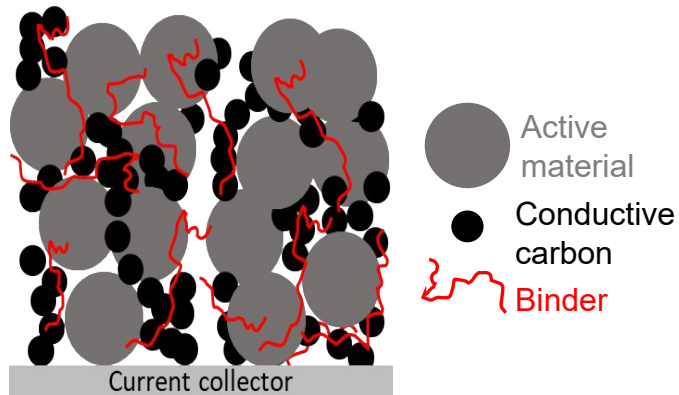


- Values given at **cell level**
- **Approx. driving ranges** for a **50 kWh** pack  
→ with EPA/WLTP cycle  
↔ close to reality



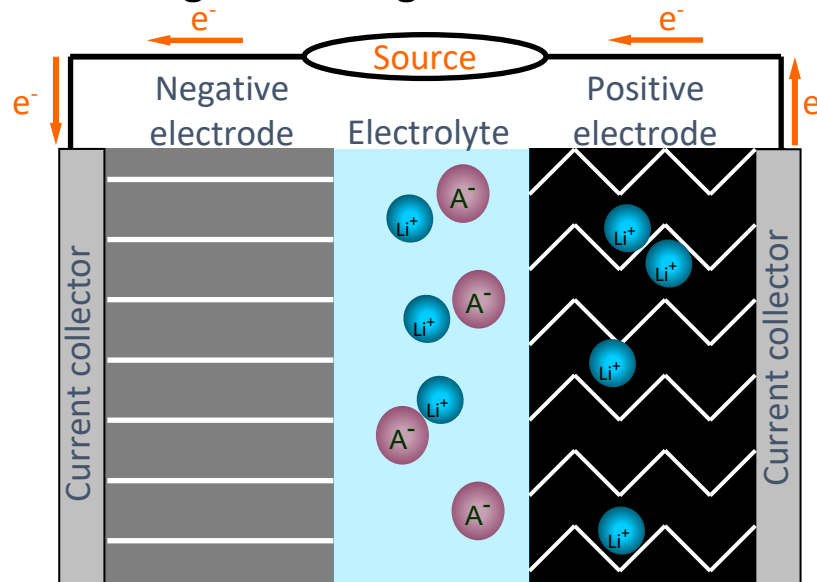
# Inside metal-ion batteries

=> Electrodes = composites



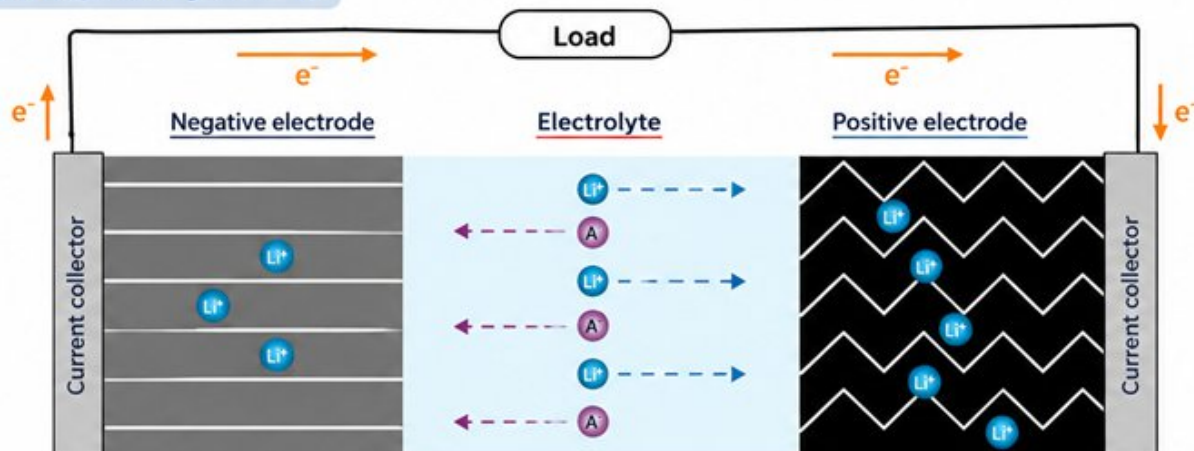
ANR seminar edition Stock-e 2009 – CALICE projet

=> M-ion charge/discharge



# Inside metal-ion batteries

## DISCHARGE (powering a device)



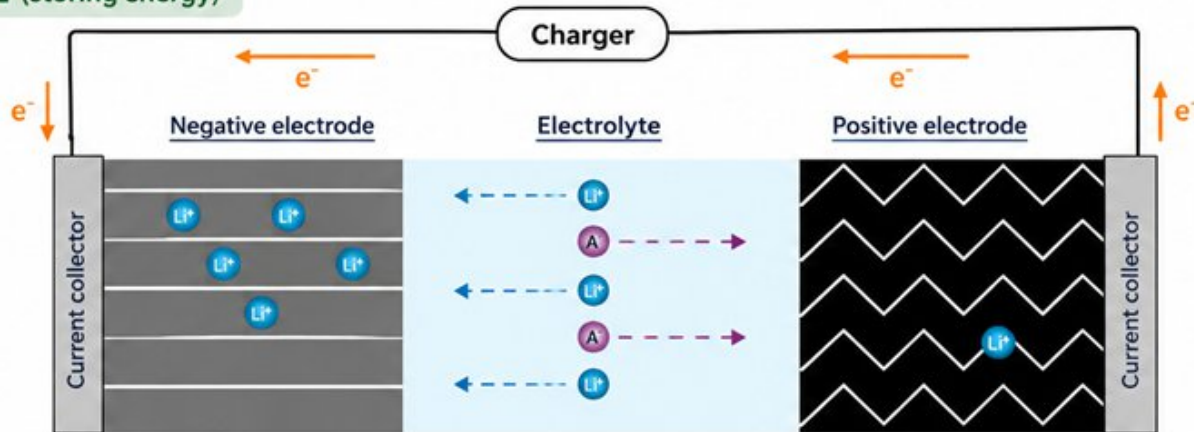
### DISCHARGE

Electrons flow from negative to positive electrode through the external circuit.

$\text{Li}^+$  Lithium ions move from negative to positive electrode through the electrolyte.

$\text{A}^-$  Anions move from positive to negative electrode.

## CHARGE (storing energy)



### CHARGE

Electrons flow from positive to negative electrode through the external circuit.

$\text{Li}^+$  Lithium ions move from positive to negative electrode through the electrolyte.

$\text{A}^-$  Anions move from negative to positive electrode.

$\text{Li}^+$  Lithium ion

$\text{A}^-$  Anion (e.g.,  $\text{PF}_6^-$ )

Graphite (negative)

Layered oxide (positive)

Electron

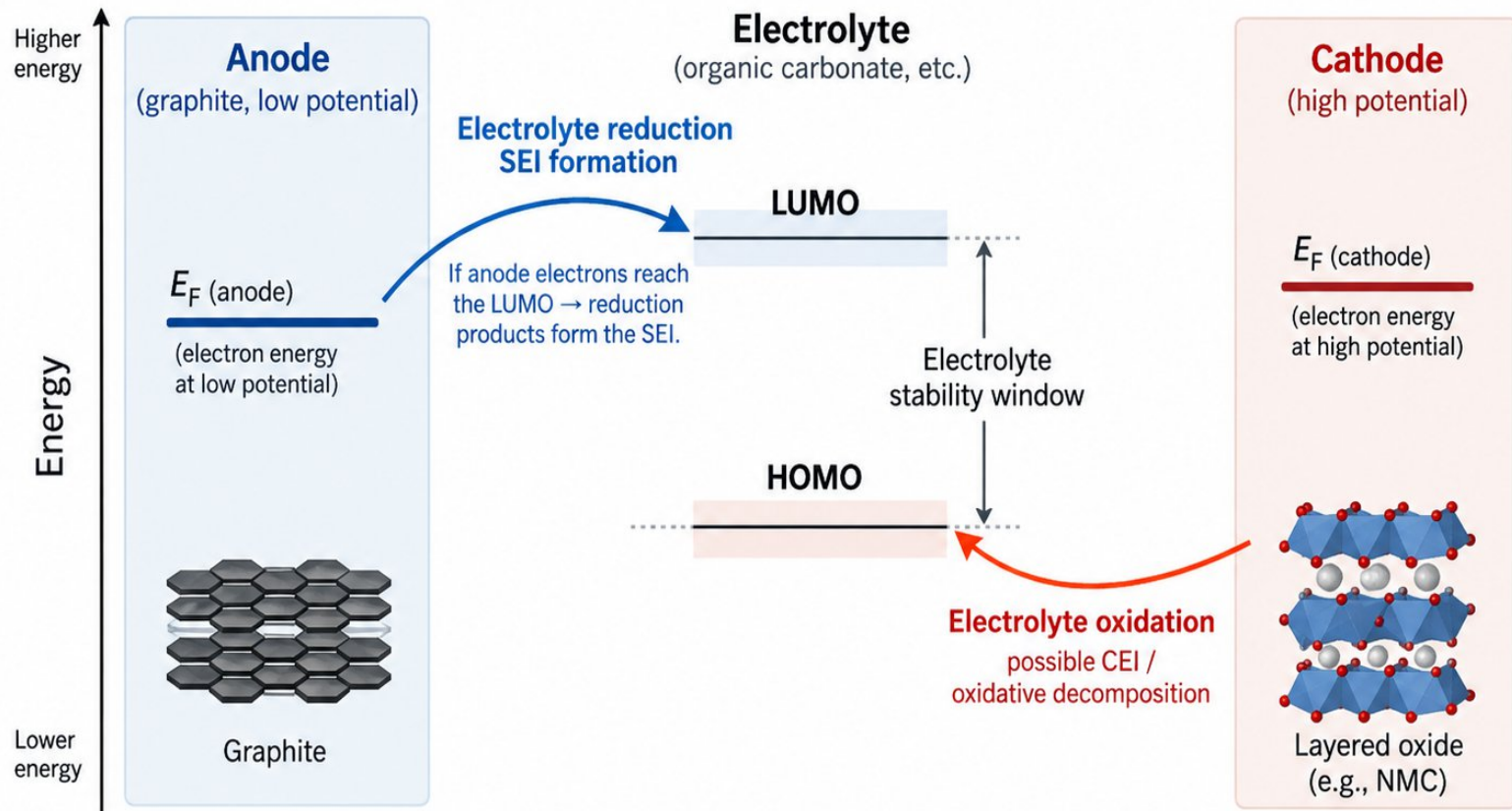
External circuit

# Outline

**2° The SEI formation and its properties**

# The Solid Electrolyte Interphase (SEI) formation

=> The HOMO-LUMO

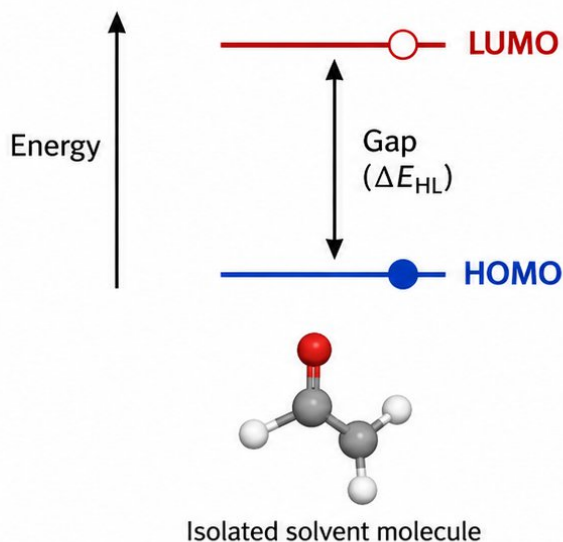


=> Ok but too simple

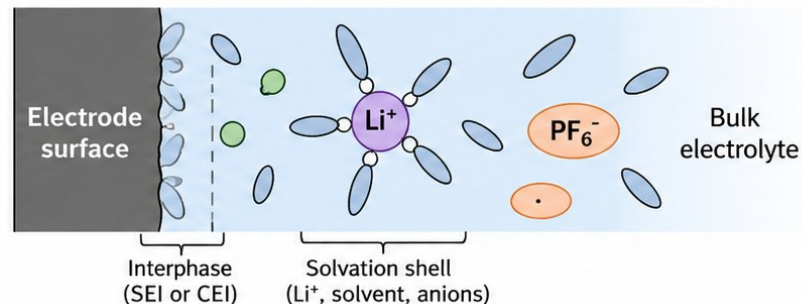
# The Solid Electrolyte Interphase (SEI) formation

## => Limits of the HOMO–LUMO

### 1. Simple HOMO–LUMO picture (isolated molecule, in vacuum)



### 2. Real battery interface (electrolyte, salt, surface, environment)



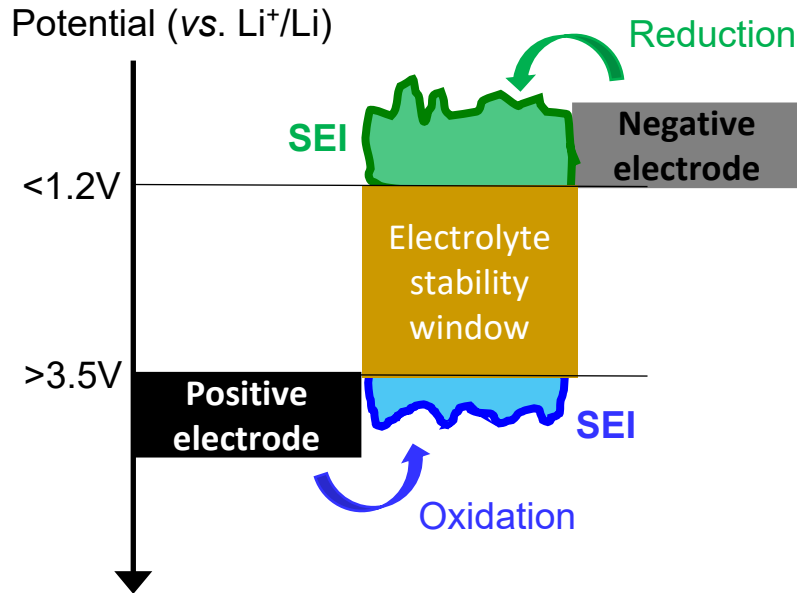
#### Key factors not captured by HOMO–LUMO

- **Solvation and ion pairing** ( $\text{Li}^+$ , anions, concentration)
- **Electrode surface** and its state of charge
- **Interfacial reactions and kinetics** (not only thermodynamics)
- **Formation of SEI / CEI** (passivation)
- **Complexity of electrolyte** (solvents, salt, additives, impurities)
- Especially problematic for **oxidation** at the **positive electrode** (surface-driven and chemical reactions)

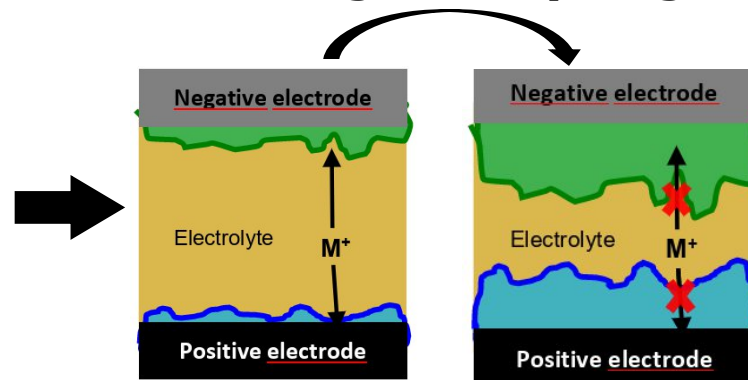
⇒ **HOMO–LUMO gap  $\neq$  electrochemical stability window**  
Useful for a qualitative picture, but not a quantitative predictive model.

# The Solid Electrolyte Interphase (SEI) formation

=> Overall: the electrolyte stability

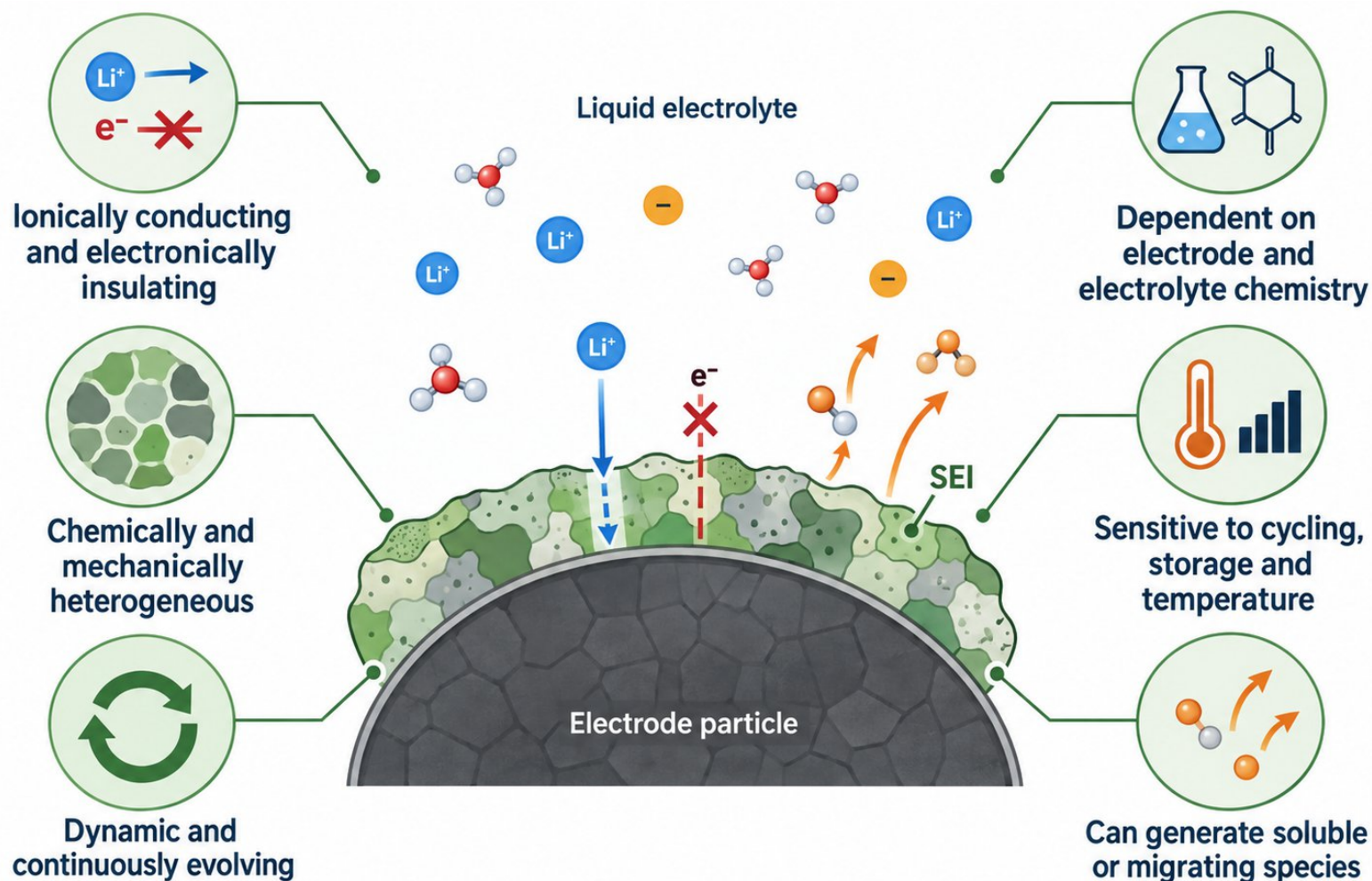


=> long term cycling



→ **SEI (Solid Electrolyte Interphase)**  
tunes the performances  
impedance = charging rate, safety, lifetime...

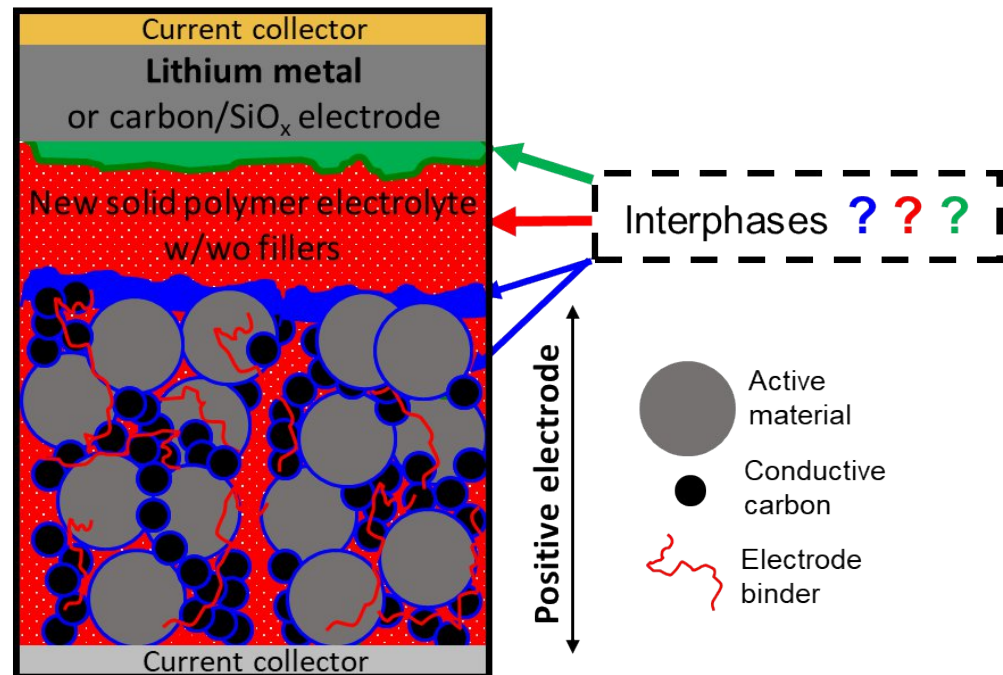
# The SEI properties



# The SEI in solid-state batteries

=> Interfaces : they are buried

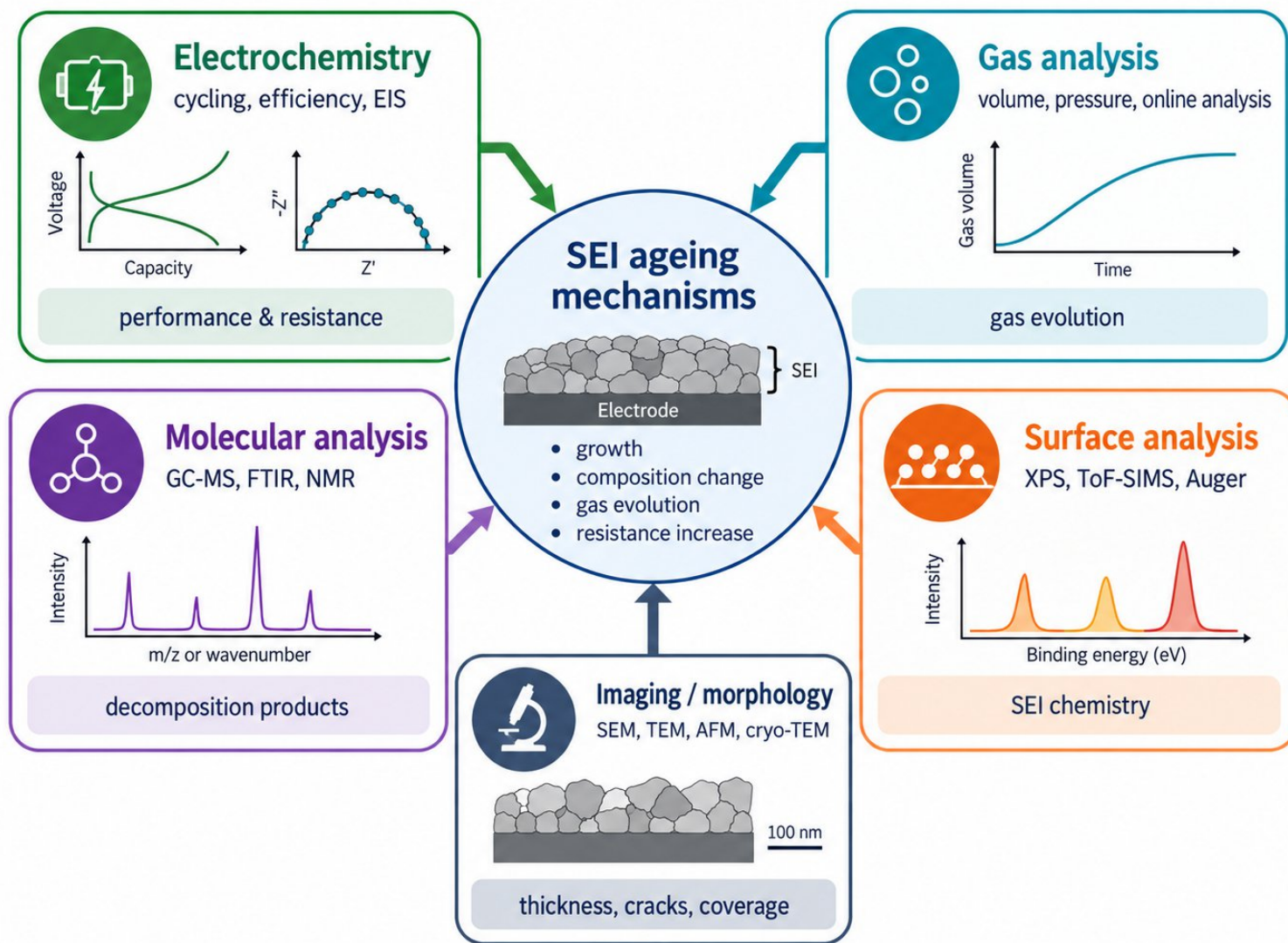
Buried solid/solid interfaces remain particularly challenging to prepare, access and analyse



# Outline

## 3° Sample preparation and analysis

# How to analyze the SEI ?



**GOAL: better understanding of parasitic phenomena at electrode/electrolyte interfaces**



to be completed

# Gas and Molecular analysis



to be completed



to be completed

# Surface and interface analysis



## Auger

Elemental mapping  
+ electrode/particles  
morphology

- High spatial resolution
- Surface sensitive

## XPS

Chemical environment  
mapping  
*(needs enough counts)*

- Chemical state information
- Lower spatial resolution

## ToF-SIMS

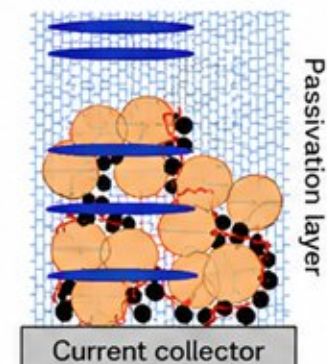
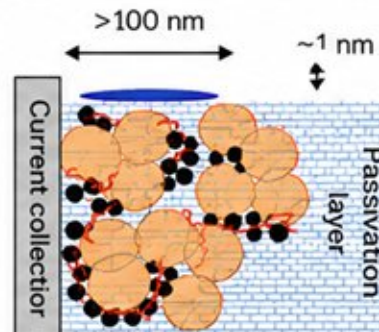
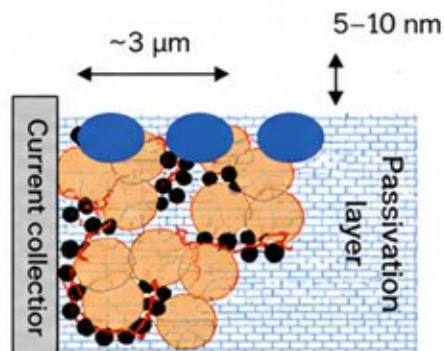
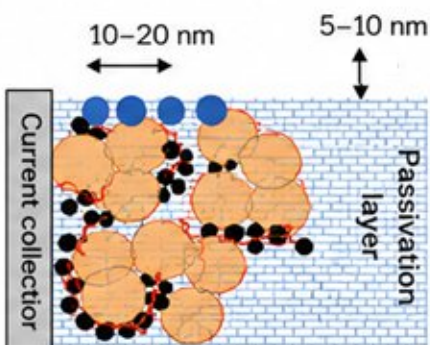
Elemental / molecular mapping

*ex-situ*  
cross-section

Lateral information  
on cross-section

*in-situ (in the machine)*  
cross-section  
(ion etching / FIB)

Depth profiling  
(through-plane)





to be completed



to be completed

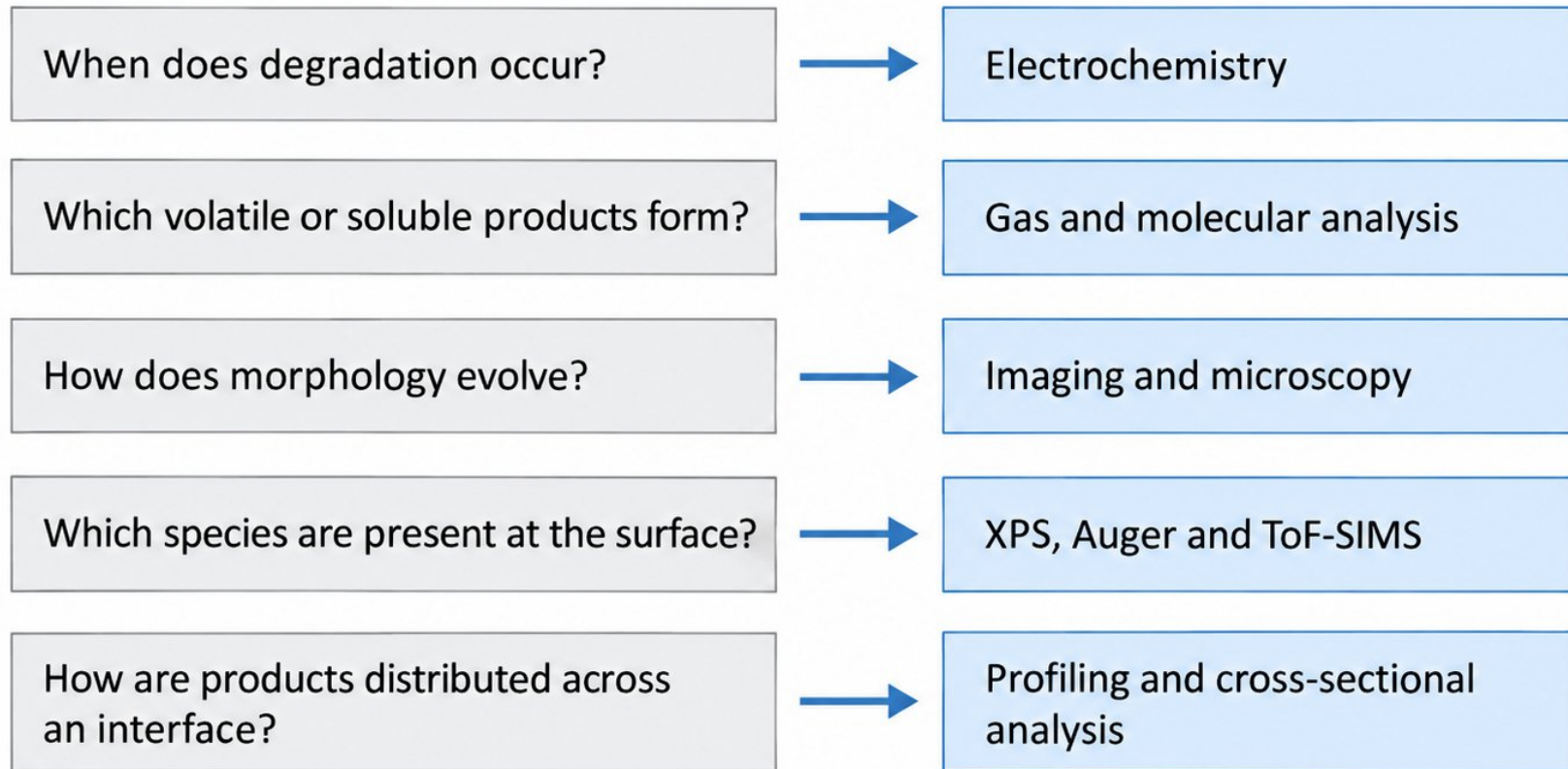


to be completed



to be completed

# Selecting a method according to the scientific question



**Complementary methods are often required**

***Thanks***

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